About Dataset

Context

Heart Disease is among the most prevalent chronic diseases in the United States, impacting millions of Americans each year and exerting a significant financial burden on the economy. In the United States alone, heart disease claims roughly 647,000 lives each year — making it the leading cause of death. The buildup of plaques inside larger coronary arteries, molecular changes associated with aging, chronic inflammation, high blood pressure, and diabetes are all causes of and risk factors for heart disease.

While there are different types of coronary heart disease, the majority of individuals only learn they have the disease following symptoms such as chest pain, a heart attack, or sudden cardiac arrest. This fact highlights the **importance of preventative measures and tests** that can accurately predict heart disease in the population prior to negative outcomes like myocardial infarctions (heart attacks) taking place.

The Centers for Disease Control and Prevention has identified high blood pressure, high blood cholesterol, and smoking as three key risk factors for heart disease. Roughly half of Americans have at least one of these three risk factors. The National Heart, Lung, and Blood Institute highlights a wider array of factors such as Age, Environment and Occupation, Family History and Genetics, Lifestyle Habits, Other Medical Conditions, Race or Ethnicity, and Sex for clinicians to use in diagnosing coronary heart disease. Diagnosis tends to be driven by an initial survey of these common risk factors followed by bloodwork and other tests.

Content

The Behavioral Risk Factor Surveillance System (BRFSS) is a health-related telephone survey that is collected annually by the CDC. Each year, the survey collects responses from over 400,000 Americans on health-related risk behaviors, chronic health conditions, and the use of preventative services. It has been conducted every year since 1984. For this project, I downloaded a csv of the dataset available on Kaggle for the year 2015. This original dataset contains responses from 441,455 individuals and has 330 features. These features are either questions directly asked of participants, or calculated variables based on individual participant responses.

This dataset contains 253,680 survey responses from cleaned BRFSS 2015 to be used primarily for the binary classification of heart disease. Not that there is strong class imbalance in this dataset. 229,787 respondents do not have/have not had heart disease while 23,893 have had heart disease. The question to be explored is:

1. To what extend can survey responses from the BRFSS be used for predicting heart disease risk?

and

2. Can a subset of questions from the BRFSS be used for preventative health screening for diseases like heart disease?

Acknowledgements

It it important to reiterate that I did not create this dataset, it is just a cleaned and consolidated dataset created from the BRFSS 2015 dataset already on Kaggle. That dataset can be found here and the notebook I used for the data cleaning can be found here.

Inspiration

Let's build some predictive models for for heart disease.